

Growing use of microcontrollers

Microcontrollers were primarily used in automobiles for the fuel injection system. But, later these found application in many other areas.

Ramesh M., senior engineer-field service, Lucas TVS, explains, "For optimal combustion of fuel in the engine, fuel quantity and timing of injection are determined with the help of various sensors for throttle position, camshaft/crankshaft position, airflow, engine RPM, rail pressure, fuel pressure, load value, air temperature, engine temperature, oxygen, vehicle speed and AC system refrigerant."

He adds, "To process these parameters, microcontrollers are incorporated as part of the fuel injection system in the terminology ECU (electronic control unit) by Bosch and ECM (engine control module) by Delphi."

Microcontrollers also find use in vehicle control modules like electronic power steering systems, adaptive cruise control systems, airbag control systems, electronic stability control systems and anti-lock braking systems. Then, there is body control unit. Its main role is to take care of the seating control unit, wiper control, power windows, central locking and door controls in cars.

of vehicle tracking, GPS, for instance, when used in the simplest form, provides the user with the ability to navigate from point to point. Building on such simple applications, consider the use of digital GPS technology in fleet management, where each vehicle can be tracked easily. This provides the authorities an added means to track the location of the vehicle."

Tan adds, "Electronics can also help to reduce the ever increasing problem of vehicular theft. Consider the implementation of such GPS units as part of the insurance coverage plan. In the first instance of car loss or an accident, the insurance company will be able to pinpoint the exact location of the vehicle and thus enlist the authorities for help."

Anti-theft systems. Car alarms are the most popular way of preventing theft. There are different locations where car alarms can be installed. Different forms of sensors allow the possibility of theft to be detected from different points in the car. Some of the popularly used sensors include car-alarm door sensor, car-alarm shock sensor, car-alarm windows and pressure sensor, and car-alarm motion and tilt sensor. The car alarm, in its simplest form, is nothing but one or more sensors connected to some sort of alarm.

Today's car alarms are much more sophisticated as these contain an array of sensors including the pressure sen-

sor and the motion sensor. A siren of distinct sound could be used to identify a car. An auxiliary battery is used so that the alarm operates even when the main battery is off.

Car misbehaviour management systems. Do cars misbehave?—you may ask. Well, that's the simplest name engineers could attach to problems that arise when the tyres of your car get punctured or your car skids on a slippery surface.

In order to control the misbehaviour of the car, real-time sensing must be done. This is done with various electronic components, placed at the right locations inside the car, which could monitor the various parts of the car. The areas that are typically monitored include the tyres, engines and brakes. In order to monitor the tyres, pressure sensors are placed on them. These sense the pressure and make the data available on the display screen of the dashboard.

Another 'misbehaviour' could be heating-up of the engine. This is monitored by having temperature sensors placed onto the engine, which output signals to the computer unit and the dashboard displays the temperature of the engine.

Personal safety systems. As indicated by the name, here the goal is to use technology for personal safety of the passengers. Primarily, these are warning systems, and sometimes they

go to the extent of restraining the car from moving. The most common example is the seat-belt warning system, which is now available in most Indian cars too. Herein, a sensor placed in the seat belt's clip alerts the driver if the seat belt is not worn. Another example is door sensors, which alert the driver if any of the doors is open.

More sophisticated electronics comes into play in airbag systems. Airbags provide additional safety along with seat belts. An airbag consists of the airbag central unit (ACU), which monitors a number of sensors like accelerometer, impact sensor, door sensor and wheel sensor. Whenever the threshold is reached, the ACU triggers the ignition of the gas generator to rapidly inflate the nylon airbag. As the vehicle occupant collides with the bag, it squeezes the bag and the air escapes through the vent holes.

Accident-avoidance systems. The basic ones alert the driver of possible threats like sharp bends or steep inclines. Advanced systems being worked upon include auto-pilot systems, where the cars would be automatically driven, and with the help of sensors and central GPS mapping systems, chances of collisions or crashing into road dividers would be avoided. Vehicle dynamics integrated management (VDIM) is an attempt in this direction.

As indicated by its name, VDIM is a system that integrates steering, braking, vehicle stability and traction control. It acquires data from the sensors placed throughout the car. The information picked up includes wheel rotation, brake pressure and movement of the car body.

Previously, braking, steering and vehicle stability, all worked independently. But in VDIM, all of it is integrated. On slippery roads, the car takes over and automatically controls steering torques to prevent a spinout. Similarly, if the driver over-steers while cornering, causing the tyre to slip laterally, VDIM takes necessary corrective measures.